



NATIONAL TEACHERS COLLEGE

Software Design & Analysis Document (SDAD) NTCentinel

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Course: I.T. Elective I



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1. Introduction

Project Overview

The NTCentinel system is an automated, kiosk-based terminal engineered to transform and digitize the manual logging of student behavioral and compliance violations at the primary entrance of National Teachers College. Historically, security guards deployed at campus perimeters relied heavily on traditional pen-and-paper logs to document entry infractions. This archaic methodology introduced operational delays, verification errors, and multi-encoding backlogs between gate operations and the backend administration.

By establishing real-time digital synchronization, NTCentinel builds an immediate network bridge between the campus security gates and the central Office of Student Affairs (OSA) database infrastructure. The system leverages high-speed identity extraction to authenticate student profiles, log infraction details automatically, trigger policy enforcement conditions, and optimize operational workflow efficiency for institutional personnel.

Chosen UN SDG Target

The implementation blueprint of the NTCentinel system directly addresses the indicators of United Nations Sustainable Development Goal 11: Sustainable Cities and Communities, specifically Target 11.7, which mandates universal provision and access to safe, inclusive, and accessible public spaces. Prior to system deployment, manual gate check-ins induced severe foot-traffic bottlenecks during peak morning hours. This delay caused crowds of incoming students to overflow into public sidewalks and exposed city streets, presenting immediate physical and traffic hazards. By driving identity validation down to sub-3-second timeframes, NTCentinel eliminates sidewalk clustering, clears institutional borders, mitigates vehicle-pedestrian hazards, and fosters a safe, secure, and optimized physical ecosystem for the immediate community.

2. Requirements Analysis

Functional Requirements

- **REQ-01 (High-Speed Decoding Matrix):** The system terminal must automatically capture and interpret encoded student QR data layouts within a maximum processing latency of 1.0 second from initial physical scanner proximity.
- **REQ-02 (Identity Verification Logic):** Upon data interpretation, the application layer must query corresponding repository states and display the associated student's digital profile photo, full legal name, and cumulative violation metrics on the guard interface.



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- **REQ-03 (Digitized Selection Interface):** The system interface must display an interactive touchscreen layout presenting a standardized, predefined category tree of school infractions, removing any requirement for arbitrary manual textual data entry by guard operators.
- **REQ-04 (Automated Infrastructure Sync):** Upon guard operational confirmation of a violation entry, the system must establish an immediate transaction log and push updates to the centralized administrative database concurrently.
- **REQ-05 (Automated Flagging & 3-Strike Rule):** The software architecture must actively process cumulative individual violation counts. If a record hits the defined threshold of three active infractions, the system must generate a flag state and dispatch a critical notification event to the OSA dashboard.
- **REQ-06 (Violation Slips Generation):** The terminal software must drive attached hardware components to automatically print a serialized physical infraction slip via an integrated thermal printing module immediately upon record insertion.

Non-Functional Requirements

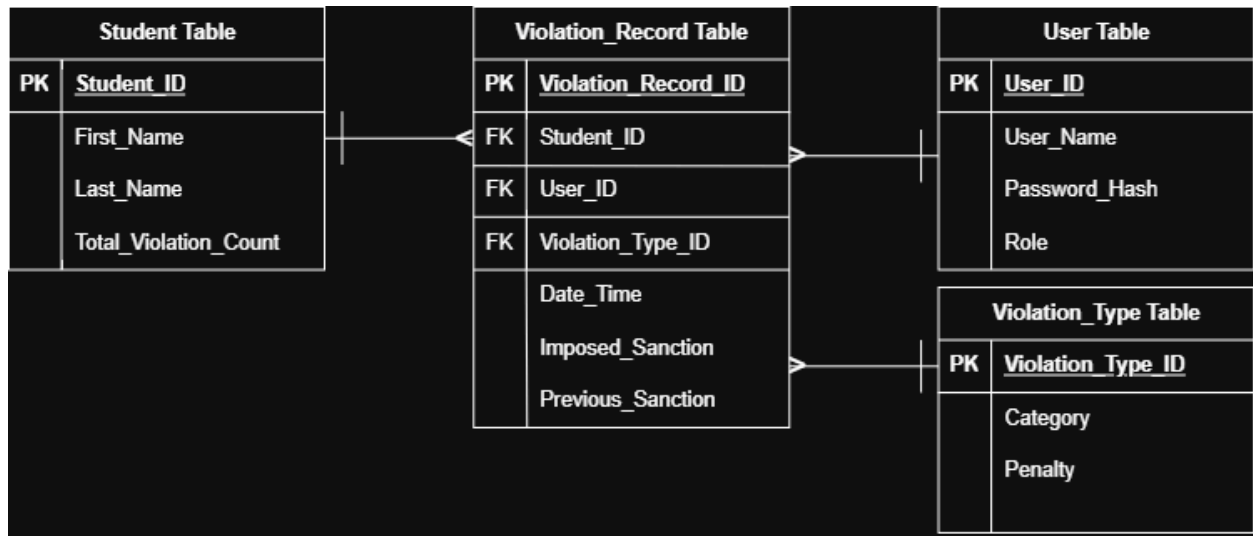
- **Performance and Throughput (REQ-07):** To completely prevent pedestrian traffic congestion at entry checkpoints, the end-to-end processing execution time—spanning initial scanning, user profile display, infraction selection, and remote write verification—must finish inside a maximum duration of 3 seconds.
- **Security and Authentication (REQ-08):** Access paths into administrative reporting parameters, master configurations, and dashboard logs are strictly restricted to authenticated OSA staff members using cryptographically secure role-based authorizations and password hashes.
- **Data Privacy Compliance (REQ-09):** System storage structures, transport pathways, and endpoint nodes must fully comply with the legal protections defined in Republic Act No. 10173 (Data Privacy Act of 2012) to safeguard confidential student information against leakages or unauthenticated visibility.
- **Network Fault Tolerance (REQ-10):** The terminal execution architecture must employ robust edge-caching capabilities. If local area network (LAN) access drops, the kiosk software must cache transaction layers safely on local machine hardware and perform bulk synchronization once host connection integrity re-establishes.
- **Hardware Usability and Environmental Factors (REQ-11):** The outdoor deployment kiosk must utilize high-luminance, high-contrast display components optimized for open-air sunlight viewability, supported by large layout interactive buttons to prevent input mistakes by security personnel.



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3. Database Design

Entity Relationship Diagram (ERD) Overview



Data Dictionary

Table 3.1: Student Table

Field Name	Data Type	Constraints	Description
Student_ID	VARCHAR(15)	PRIMARY KEY, NOT NULL	Unique school ID mapped directly from the student card QR code scan.
First_Name	VARCHAR(50)	NOT NULL	Given names of the student profile.
Last_Name	VARCHAR(50)	NOT NULL	Family surname of the student profile.



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Total_Violation_Count	INT	DEFAULT 0, NOT NULL	Real-time running sum tracking the active 3-strike rule limit.
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Table 3.2: Violation_Record Table

Field Name	Data Type	Constraints	Description
Violation_Record_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Internal unique auto-generated reference integer for every single logged event.
Student_ID	VARCHAR(15)	FOREIGN KEY, NOT NULL	Establishes direct link to target entity in Student Table.
User_ID	INT	FOREIGN KEY, NOT NULL	Establishes direct link to logging operator in User Table.
Violation_Type_ID	INT	FOREIGN KEY, NOT NULL	Establishes link to policy parameters in Violation_Type Record.
Date_Time	DATETIME	DEFAULT CURRENT_TIMESTAMP	System timestamp capturing exact execution date/time.
Imposed_Sanction	VARCHAR(100)	NULL	The specific active penalty issued based on current offense repetition.



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Previous_Sanction	VARCHAR(100)	NULL	Auditable record tracking historical past status for baseline reference.
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Table 3.3: User Table

Field Name	Data Type	Constraints	Description
User_ID	INT	PRIMARY KEY, NOT NULL	Unique surrogate identifier for institution security or admin employees.
User_Name	VARCHAR(30)	UNIQUE, NOT NULL	The unique access handle/username utilized for application authentication.
Password_Hash	VARCHAR(255)	NOT NULL	Secure salted cryptographic string mapping authorization passwords.
Role	VARCHAR(20)	NOT NULL	System privileges control group (e.g., 'Security Guard', 'OSA Staff').



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Table 3.4: Violation_Type Record

Field Name	Data Type	Constraints	Description
Violation_Type_ID	INT	PRIMARY KEY, NOT NULL	Predefined primary lookup index for standard institutional infractions.
Category	VARCHAR(50)	NOT NULL	The textual description of offense groupings (e.g., 'Uniform Policy').
Penalty	VARCHAR(100)	NOT NULL	Standard legal or academic consequence tied to this specific infraction row.



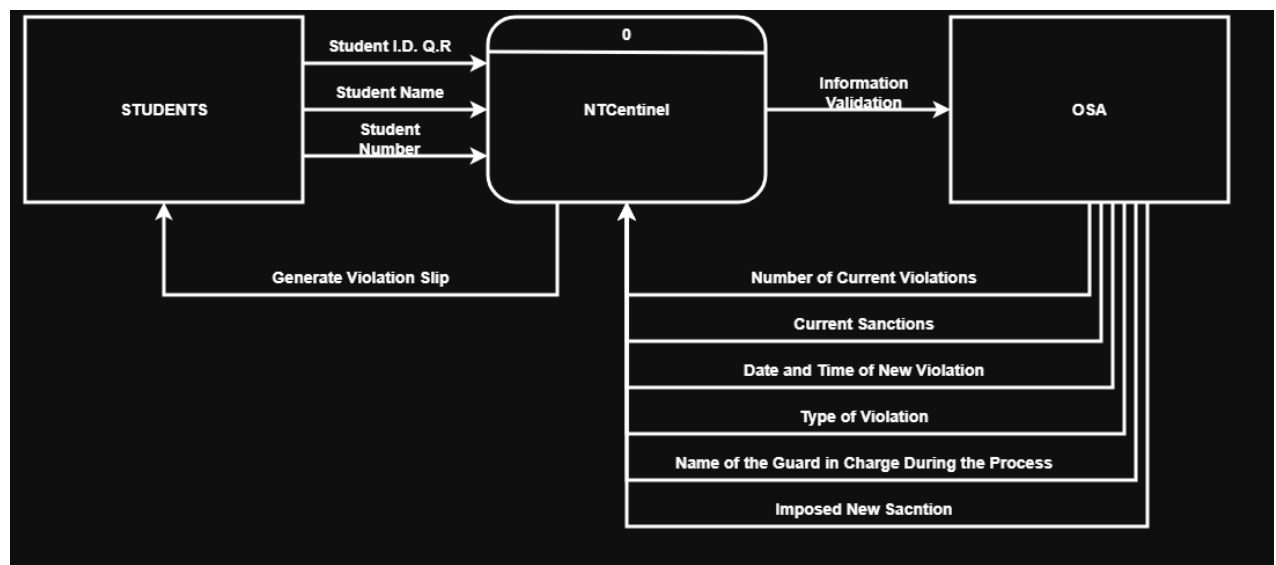
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4. System Architecture

Data Flow Diagram (DFD) Level 0: Context View

The high-level context architecture traces processing paths between the centralized NTCentinel boundary and external entities:

- **Student to System Input:** Initiated via direct physical matrix presentation, transmitting structural attributes: Student I.D. QR, Student Name, and Student Number into the kiosk reception subsystem.
- **Kiosk to Backend OSA Exchange:** Handles transaction pipelines over secure network links, delivering Information Validation payloads. The backend returns live state structures to the kiosk UI including: Number of Current Violations, Current Sanctions, Date and Time of New Violation, Type of Violation, Name of the Guard in Charge, and the specific calculated Imposed New Sanction.
- **System to Student Output:** Upon internal data validation and commit confirmation, the terminal signals local hardware to issue a tangible Generated Violation Slip directly to the student.





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Data Flow Diagram (DFD) Level 1: Process Decomposition

The logical architecture scales down the operational boundary into four primary isolated data sub-processes and two system data repositories:

- **Process 1.0 (Process Information):** Consumes hardware QR input sequences, abstracts raw strings into standard identifiers, issues search keys into the Student Database, and intercepts returned Student Record strings to establish profile context.
- **Process 2.0 (Log Violation):** Active upon identity confirmation. This component exposes the interactive selection panel and resolves logic across isolated operations: checking the Number of Current Violations (2.1), assessing Current Sanctions (2.2), executing Capture Timestamp (2.3), confirming the Type of Violation (2.4), attaching the Name of Guard in Charge (2.5), and logging the updated New Sanction (2.6) before submitting a complete Violation Record payload down into the Violation Database storage unit.
- **Process 3.0 (Track Violation):** Evaluates database append operations in real-time, executing checks on the 3-strike rule count and refreshing parameters to keep user records updated.
- **Process 4.0 (Generate Violation Report):** Monitors infraction flags. When limit rules breach, it triggers immediate administrative alerts to the OSA Dashboard interface and controls localized peripheral tasks to handle immediate thermal receipt print out actions.

